

kerosene-run engines consumed more fuel and polluted the water, thus affecting the marine life adversely. With conventional in-board marine diesel engines, beach landings were also very difficult. After rigorous research, Mohanlal developed in 2001 a gearbox and manually tiltable drive system for small-capacity diesel engines to overcome these problems. His 'Z drive system' can be locked at 90 de-

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grees while landing. The gear box has been developed for a 12HP diesel engine with 2,400 RPM idling speed and around 12 litres fuel tank capacity. Mohanlal's inventiveness was on best display in gearbox design, using helical gears, reworking the clutch system to enable smooth running and better power transmission for the diesel engines.

But as Peter Drucker said technology is not about tools but about how man works, Mohanlal discovered that excess weight was a problem. "Since it is not possible to leave the engine in fishing boats, one has to

carry it home at the end of the day. So reducing the load was my next big challenge," he says. In 2007, Mohanlal did so by using aluminum alloys instead of steel. But the speed of the boat was only 7-9 km per hour. So increasing speed was next on his agenda. "Invention makes you vulnerable. It's like a disease that makes you restless each moment of your life until you reach the point of perfection," he says in a matter-of-fact way. He went on improvising until he developed a gearbox which ultimately got approval from Matsyafed, the Kerala State Cooperative Federation for Fisheries Development.

In June 2008, he could increase the speed to 20 km per hour. Since Matsyafed was spending money on kerosene subsidy, they realised the potential of this innovation and started promoting it. Now Matsyafed has come forward to commercialise his innovation and has ordered 25 gearboxes from Mohanlal. His fame crossed the boundaries of Kerala when the National Innovation Foundation honoured him with an award for Grassroots Unaided Technological Innovations and Outstanding Traditional Knowledge in September 2009. The foundation has also allowed Mohanlal to network with the South India Fisheries Federation and promote the technology. Using a gear-powered engine cuts down fuel costs by two thirds. With the gearbox one needs to pay only Rs 22 for an hour's work while the regular one costs Rs 110 for the same work.

It also minimises pollution with a projected cost of less than Rs 60,000. In Kerala alone there are more than 20,000 boats which may directly benefit from this innovation. In order to dedicate time to his innovations, Mohanlal had to leave his fishing boat workshop which earned thousands of rupees per day, but he says has never regretted the move.

Determination and perseverance count. Mohanlal may not have studied beyond Class X but his strong practical vision is a textbook lesson in the effectiveness of a great idea. ■

Bhargavan MOHANLAL, 54

INNOVATION

Gearbox for marine diesel engines

SELLING PRICE

Rs 1.35 lakh

WHAT IT DOES

Made of light aluminium alloys, it reduces the weight of the engine and leads to fuel savings

THE NEW GEARBOX HAS REDUCED THE RUNNING COST FROM Rs 110 TO Rs 22 PER HOUR.

MOHANLAL ON A BOAT WITH AN IMPROVED DIESEL ENGINE